A 4×4 Obstruction to Polynomial Ordered Binary Matrix Removal

Sparse zero localization despite unpinned anchors

Research record prepared in this investigation

9 October 2026

Abstract

The strongest prior result located in this investigation is an October 7 ProveIt research manuscript giving a 5×5 obstruction, improving OpenAI's 66×66 construction. That manuscript also displays a 4×4 candidate and an unpinned copy that survives its proposed repair. We prove that this previously displayed candidate does fail polynomial ordered binary matrix removal in the original two-group host. The new structural fact is that every copy, including every unpinned copy, maps its prescribed zero at coordinate $(3, 4)$ to a leaf row and its parent-minus column block. The order of the two minus-parity modes excludes the only possible nonterminal exception.

For $h \geq 1$, put $m = 2^h$, $d_h = 14h + 2$, and $n_h = d_h m$. The explicit hosts have normalized edit distance at least d_h^{-2} from H_4-freeness under arbitrary binary changes. Every copy meets a fixed $2m$-cell set at a specified pattern coordinate, so $N_{H_4}(A_h) \leq 2mn_h^6$. A complete three-family classification further gives $N_{H_4}(A_h) = (41m^6 - 74m^5)/4$ for $h \geq 2$, and $N_{H_4}(A_1) = 128$.

All proofs are included. Two separately implemented exhaustive programs agree on the complete weighted copy sets through depth five. Their finite checks support the implementations; they do not replace the all-depth proof. The new theorem has not been kernel-checked in Lean. No minimality or exhaustive publication-priority claim is made.

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1 Strongest baseline, contribution, and precise statement

1.1 Dated prior-work finding

As checked on 9 October 2026, current `openai/math` main is `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`, the snapshot supplied for this investigation. Its family-190 manuscript and Lean catalogue still specify a 66×66 counterexample [1, 2]. The public main history has three commits and one merged pull request. The family-190 paths have not changed since the initial public commit.

A stronger external result was found through GitHub code search: *A 5×5 Obstruction to Polynomial Ordered Matrix Removal*, a ProveIt research draft dated 7 October 2026 [3]. The first returned path-history commit is `bfbfe915186310fdf745a7a089e7dd88eeb773b1`, dated 7 October at 21:24:26 UTC. It supplies ordinary proofs for

$$H_5 = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}, \qquad n = (20h+2)2^h,$$

with exactly $2(2^h)^8$ copies and minimum repair size 4^h. Its main structural tool is full-pattern pinning with the crown $J_3 - I_3$. We reconstructed its principal pinning, counting, and arbitrary-edit proofs and found no gap in those statements. Its code was also rerun successfully; that rerun is distinguished from our independent implementations. The prior result is not Lean-formalized.

The same prior manuscript already displays the matrix H_4 below, and gives an unpinned copy surviving a dummy-row-by-leaf-column repair. It leaves the possibility of a different counting argument open. **The candidate matrix itself is prior work. The contribution here is proving that this candidate works, by classifying and controlling its unpinned copies.**

The first concrete dense-anchor reduction considered was a six-position dense prefix with 32 binary-word rows, giving a 38×38 anchor. That reduction is valid, as detailed in Appendix A, but is superseded by the located 5×5 theorem. The precise stronger size target therefore became 4×4.

1.2 Definitions and every quantifier

Write $[n] = \{1, \dots, n\}$. For a binary $k \times k$ matrix H and a binary $n \times n$ matrix A, an ordered copy is a pair of strictly increasing maps $r, c : [k] \to [n]$ satisfying

$$A(r(i), c(j)) = H(i, j) \quad \text{for every } i, j \in [k].$$

The maps are chosen independently. Every zero and every one must match. Rows and columns need not be consecutive. There is no quotient by permutations or other symmetries. Let $N_H(A)$ count these pairs. The matrix A is H-free exactly when $N_H(A) = 0$.

For $n \geq 1$, define

$$\operatorname{dist}_H(A) = \frac{1}{n^2} \min_{\substack{B \in \{0,1\}^{[n] \times [n]} \\ N_H(B) = 0}} \left| \{(x, y) \in [n]^2 : A(x, y) \neq B(x, y)\} \right|.$$

For our patterns the minimum is nonempty because the all-zero matrix is free. Edits can be anywhere and can change bits in either direction. The normalization of copy counts is n^{2k}; using $\binom{n}{k}^2$ instead only changes fixed-pattern constants in the relevant range.

A fixed H has polynomial ordered removal if

$$\exists c_H, C_H > 0\ \forall n \in \mathbb{N}_{\geq 1}\ \forall \varepsilon \in (0,1)\ \forall A \in \{0,1\}^{[n]\times[n]}:$$
$$\operatorname{dist}_H(A) \geq \varepsilon \implies N_H(A) \geq c_H \varepsilon^{C_H} n^{2k}.$$

Failure for the explicit H_4 means

$$\forall c, C \in \mathbb{R}_{>0}\ \exists n \in \mathbb{N}_{\geq 1}\ \exists \varepsilon \in (0,1)\ \exists A \in \{0,1\}^{[n]\times[n]}:$$
$$\operatorname{dist}_{H_4}(A) \geq \varepsilon \quad \text{and} \quad N_{H_4}(A) < c\varepsilon^{C} n^{8}.$$

The constructed sequence in fact supplies such witnesses with arbitrarily large n and arbitrarily small positive ε.

Theorem 1.1 (Four-by-four obstruction)**.** *Let*

$$\boxed{H_4 = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{pmatrix}.}$$

For every integer $h \geq 1$, set

$$m = 2^h, \qquad d_h = 14h + 2, \qquad n_h = d_h m, \qquad \varepsilon_h = d_h^{-2}.$$

The deterministic host A_h in Section 3 satisfies

$$\operatorname{dist}_{H_4}(A_h) \geq \varepsilon_h, \qquad N_{H_4}(A_h) \leq 2mn_h^6.$$

More precisely,

$$N_{H_4}(A_h) = \begin{cases} 128, & h = 1, \\ \dfrac{41m^6 - 74m^5}{4}, & h \geq 2. \end{cases}$$

Consequently

$$\frac{N_{H_4}(A_h)}{n_h^8} \leq \frac{41}{4}\varepsilon_h^4\, 4^{-h},$$

and polynomial ordered binary matrix removal fails for H_4.

2 Reconstruction of the published baseline

2.1 The 64-by-64 anchor and 2-by-2 body

For zero-based row and column indices $u, v \in \{0, \ldots, 63\}$, the published anchor is $J_{64} - I_{64}$, except that rows $32, \ldots, 63$ and columns $59, \ldots, 63$ are replaced by all five-bit words:

$$S_{64}(u, 59 + j) = \left\lfloor \frac{u - 32}{2^{4-j}} \right\rfloor \bmod 2 \quad (32 \leq u < 64,\ 0 \leq j < 5).$$

With e_1, e_2 the first two coordinate vectors, the pattern is

$$H_{66} = \begin{pmatrix} S_{64} & e_1 & e_2 \\ e_1^{\mathsf{T}} & 1 & 0 \\ e_2^{\mathsf{T}} & 1 & 1 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Rows and columns of S_{64} are distinct. Each row has at least 58 ones, each column at least 48. The first 32 positions on either axis give a dense core: any row or column has at most one zero there. The 32 word rows shatter the final five columns.

2.2 Host, distance, and count are separate ingredients

The host uses the signed binary-tree positions and six modes per level given explicitly below. For a general anchor order s, there are $6hs$ anchor groups, $2h+1$ variable classes, and one dummy class per axis, each of size $m = 2^h$. Thus

$$n = (6hs + 2h + 2)m.$$

For $s = 64$, this is $(386h + 2)m$.

The published distance proof selects a common random path and independent anchor, dummy, and within-block representatives. Every protected changed cell is encountered with probability m^{-2}. Fewer than m^2 edits therefore leave one protected selection intact. Avoiding its four types of P-body would propagate opposite root values to the same leaf cell, a contradiction. This proof applies to the edited matrix itself, including copies that edits create.

The published counting proof is different. A nonanchor variable position has at most four incidences with selected distinct anchor groups. Nonanchor rows do not shatter five nonanchor columns. The dense core and word block therefore pin every leading S_{64}-copy to one complete mode anchor. Its body then has the designated roles. Three mode types are impossible; the fourth forces a common diagonal leaf cell. There are m such cells, and hence

$$N_{H_{66}}(A_h) \le mn^{130}, \qquad \frac{N_{H_{66}}(A_h)}{n^{132}} \le (386h + 2)^{-2}2^{-h}.$$

Changing the original leaf-diagonal ones to zeros hits all original copies but creates plus-mode copies. A transversal is not a repair.

This distinction is consistent with Alon–Ben-Eliezer's Theorem 1.2, which concerns entry-disjoint existing copies, whereas their Problem 1.4 asks about genuine edit distance [4]. Qualitative ordered removal is established in [5]. The singleton polynomial question is Conjecture 4.5 in the first version of [6]. The ordered-graph classification in [7] concerns one common vertex order, a different model from independent row and column selections.

2.3 What the actual Lean implementation proves

The solution module `OAI.Combinatorics.MatrixRemoval.Main` proves `OAI.Problem348.counterexample_sequence` and `OAI.Problem348.no_polynomial_removal_bound`. The underlying definitions use two strictly increasing maps and Boolean equality at every cell. Hamming distance counts all differing cells. The minimum ranges over all binary matrices, assigning nonfree matrices the sentinel $n^2 + 1$; the all-zero free matrix makes this equivalent to the stated minimum when $n \ge 1$.

The sequence theorem is unconditional for the fixed 66×66 pattern. Its concrete host, all-copy bound, and arbitrary-repair bound are discharged by the imported implementation. The comparator file's `sorry` is a challenge placeholder, not the solution proof. Static inspection of 61 MatrixRemoval implementation files, totaling 7,609 lines, found no `sorry`, `admit`, `axiom`, or `native_decide`. The comparator permits `propext`, `Quot.sound`, and `Classical.choice`; it has `enable_nanoda=false`. The toolchain is Lean 4.34.1. **No Lean build or axiom-printer invocation was executed here.** These are source-level findings, not a newly rerun kernel certificate.

3 The explicit two-group host

3.1 Positions, multiplicities, and order

Fix $h \geq 1$ and $m = 2^h$. At depth i, nodes have indices $0 \leq p < 2^i$; write $\pi(p) = \lfloor p/2 \rfloor$. For $0 \leq i < h$, each node has disjoint row blocks $R_i^-(p), R_i^+(p)$ and column blocks $C_i^-(p), C_i^+(p)$, all of size $m/2^i$. At depth h, the signs are identified:

$$R_h^-(p) = R_h^+(p) = R_h(p), \qquad C_h^-(p) = C_h^+(p) = C_h(p),$$

and each leaf block has size one. These are the only identifications.

A variable class R_i^σ or C_i^σ is the union of all blocks at a fixed depth and sign. The shared leaf class occurs once. Every class has size m. For $i \geq 1$, a subscript $d \in \{0, 1\}$ restricts to nodes $p \equiv d \pmod 2$. Add dummy row and column blocks R_*, C_*, each of size m.

The recursive row order at a nonleaf node is its minus block, the zero-child subtree, the one-child subtree, then its plus block. The column order is its plus block, the same two subtrees, then its minus block. Place a leaf block once. Put R_* after all variable rows and C_* before all variable columns. Within each block use indices $0, \ldots, |\text{block}| - 1$ in increasing order.

For $1 \leq i \leq h$, comparisons with a parent-level block give

$$\begin{aligned} R_i^+(p) < R_{i-1}^+(a) &\iff \pi(p) \leq a, \quad R_{i-1}^-(a) < R_i^-(p) \iff a \leq \pi(p), \\ C_{i-1}^+(b) < C_i^+(q) &\iff b \leq \pi(q), \quad C_i^-(q) < C_{i-1}^-(b) \iff \pi(q) \leq b. \end{aligned} \tag{1}$$

Here $X < Y$ means every position of the first block precedes every position of the second. The formulas hold for shared leaves too. Indeed, depth-$(i-1)$ subtrees occur in increasing node order, and each child lies between its parent's opening and closing blocks. At a fixed nonleaf depth j,

$$C_j^-(a) < C_j^+(b) \iff a < b, \qquad C_j^+(a) < C_j^-(b) \iff a \leq b. \tag{2}$$

Complements of the comparisons in (1) give the strict reverse inequalities used below; the compared blocks are disjoint.

3.2 Modes and anchor groups

At every depth $i = 1, \ldots, h$, introduce the six modes, in this order:

$$V_i^+,\ V_i^-,\ W_{i,0}^+,\ W_{i,1}^+,\ W_{i,0}^-,\ W_{i,1}^-.$$

Modes at smaller depths come first. Use this same mode order on both axes. Their four role sets are

t	$\mathsf{R}_1(t)$	$\mathsf{R}_2(t)$	$\mathsf{C}_1(t)$	$\mathsf{C}_2(t)$
V_i^+	R_i^+	R_{i-1}^+	C_*	C_{i-1}^+
V_i^-	R_{i-1}^-	R_i^-	C_*	C_{i-1}^-
$W_{i,d}^+$	$R_{i,d}^+$	R_*	C_{i-1}^+	$C_{i,d}^+$
$W_{i,d}^-$	$R_{i,d}^-$	R_*	$C_{i,d}^-$	C_{i-1}^-

(3)

For each mode, the two row roles are disjoint, as are the two column roles. Correct ordering of representatives is a separate requirement; whole role sets need not precede one another.

Before every variable and dummy position, place two anchor groups per mode on each axis, denoted $\widehat{R}_t(1), \widehat{R}_t(2)$ and $\widehat{C}_t(1), \widehat{C}_t(2)$. Every group has size m; group 1 precedes group 2. Thus there are $12h$ anchor groups and $2h + 2$ nonanchor classes per axis:

$$n = (12h + 2h + 2)m = (14h + 2)m.$$

Identify the ordered axes with $[n]$.

3.3 Every entry

Between anchors set

$$A_h(\widehat{R}_t(u), \widehat{C}_{t'}(v)) = \mathbf{1}[t = t']\mathbf{1}[u \neq v].$$

A nonanchor row in $\mathsf{R}_j(t)$ has signature e_j^{T} against the two anchor columns of t; outside both roles it has signature zero. A nonanchor column in $\mathsf{C}_j(t)$ has signature e_j against its anchor rows; otherwise its signature is zero.

Between nonanchors put one whenever at least one position is dummy. All remaining nonzero entries are specified in this table:

row block	column block	range	A_h
$R_i^+(p)$	$C_i^+(q)$	$0 \le i \le h$	$\mathbf{1}[p \le q]$
$R_i^-(p)$	$C_i^-(q)$	$0 \le i < h$	$\mathbf{1}[p < q]$
$R_i^+(p)$	$C_{i-1}^+(q)$	$1 \le i \le h$	$\mathbf{1}[\pi(p) \le q]$
$R_i^-(p)$	$C_{i-1}^-(q)$	$1 \le i \le h$	$\mathbf{1}[\pi(p) < q]$

(4)

Cases absent from the table are zero. Entries are constant on a pair of blocks. The same-level minus line excludes the leaves, so the shared leaf-by-leaf threshold is weak. In the last two lines the column depth is below h, and its two signs are distinct. Thus the assignments never conflict.

4 Structural facts valid at every depth

Lemma 4.1 (Twins, nesting, and signatures)**.** *The following hold for every $h \ge 1$.*

1. *Positions within any one anchor group are full row or column twins. Positions in a dummy group are full twins too.*
2. *An anchor position's positive nonanchor neighborhood is contained in one variable class, a parity subclass, or the dummy class.*
3. *Within a variable row class, traces on opposite nonanchors are nested by inclusion as the node index increases. The same statement holds for variable columns. Within any variable class there are at most two full traces on opposite anchors.*
4. *For a fixed* nonanchor *row, its entries along a variable column class are nondecreasing in node order. A nonleaf variable row can be nonzero only in its own signed column class and its signed parent class. A leaf row can meet only the leaf class and the two signed parent classes.*
5. *A row in $\mathsf{R}_2(t)$ and a column in $\mathsf{C}_1(t)$ always have entry one.*

Proof. The first two statements follow from group-constant entries and the role table. The variable thresholds in (4) are nonincreasing in the row node and nondecreasing in the column node. Their zero patterns depend only on the class; dummy incidences are constant. This proves nesting and the nonanchor-row monotonicity assertion. Anchor rows themselves need not have monotone variable traces, because their incidences can select parity. The stated restriction is essential.

For completeness, the variable mode incidences, with undefined levels omitted, are as follows; (t, j) means role j:

$R_i^+(p)$, $i < h$	$(V_i^+, 1), (V_{i+1}^+, 2), (W_{i,d}^+, 1)$
$R_i^-(p)$, $i < h$	$(V_i^-, 2), (V_{i+1}^-, 1), (W_{i,d}^-, 1)$
$R_h(p)$	$(V_h^+, 1), (V_h^-, 2), (W_{h,d}^+, 1), (W_{h,d}^-, 1)$

and

$$\begin{array}{c|l} C_i^+(p),\ i<h & (V_{i+1}^+,2),(W_{i+1,0}^+,1),(W_{i+1,1}^+,1),(W_{i,d}^+,2) \\ C_i^-(p),\ i<h & (V_{i+1}^-,2),(W_{i+1,0}^-,2),(W_{i+1,1}^-,2),(W_{i,d}^-,1) \\ C_h(p) & (W_{h,d}^+,2),(W_{h,d}^-,1). \end{array}$$

Here $d = p \bmod 2$. Only parity varies within a fixed class, proving the two-signature bound, including at the shared leaves. These lists also verify the four-mode incidence bound used in the original dense anchor argument. Finally, in a V-mode column role 1 is dummy, and in a W-mode row role 2 is dummy, proving the last assertion. □

Lemma 4.2 (Nonanchor crown exclusion)**.** *The nonanchor submatrix contains no row-and-column permutation of* $\mathcal{C}_3 = J_3 - I_3$.

Proof. Such a crown would have three distinct two-element row traces on three columns. A dummy column is impossible because it is one on every nonanchor row. Two columns from the same variable class are impossible because their traces are nested, whereas a crown has incomparable column traces. Thus the columns occupy three distinct variable classes. A dummy row has trace 111, so cannot supply a crown row.

Join two column classes whenever the support of some variable row can meet both. This graph is *contained in* the graph formed by two signed paths

$$C_0^\sigma - C_1^\sigma - \cdots - C_{h-1}^\sigma \quad (\sigma = +, -)$$

and the terminal triangle

$$T = \{C_{h-1}^+, C_{h-1}^-, C_h\}.$$

The containing graph has only this triangle. Some displayed edges need not be realized at small depths; equality of graphs is not used. The three required two-element supports would force the selected classes to be T. Every variable row able to meet two classes of T is a leaf row. Its trace is a permutation of

$$(\mathbf{1}[\pi(p) \le q_+], \mathbf{1}[\pi(p) < q_-], \mathbf{1}[p \le q_0]),$$

which is nested as p increases. It cannot supply three incomparable two-element traces. This proves the lemma, including $h = 1$. □

5 Every ordered copy localizes to a sparse zero set

Theorem 5.1 (All-depth sparse zero localization)**.** *In every ordered* H_4*-copy in* A_h*, its third row is a leaf row* $R_h(p)$*, and its fourth column belongs to* $C_{h-1}^-(\pi(p))$*. Thus the prescribed zero at pattern coordinate* $(3,4)$ *lies in*

$$Z_h = \bigcup_{p=0}^{m-1} R_h(p) \times C_{h-1}^-(\pi(p)), \qquad |Z_h| = 2m.$$

5.1 Exhaust all possible anchor counts

Lemma 5.2. *Every copy has exactly two anchor rows and either one or two anchor columns. With two anchor columns, both axes use the two complete groups of one mode and the other positions have that mode's designated roles.*

Proof. Let a, b be the total numbers of anchors among its four rows and four columns. They are prefix counts because all anchors come first. All four rows of H_4 are distinct, as are all four columns.

Suppose $a \le 1$. If $b = 0$, rows $2, 3, 4$ and columns $1, 3, 4$ are nonanchors and form

$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix},$$

a permuted crown, contradicting Lemma 4.2. If $1 \le b \le 2$, the first anchor column has ones in nonanchor rows 2 and 3, putting them in one nonanchor class. On nonanchor columns 3 and 4 their traces are 01 and 10, impossible for nested variable traces or dummy twins. If $b \ge 3$, anchor column 1 connects rows 2 and 3, and anchor column 3 connects rows 3 and 4. The three nonanchor rows consequently lie in one class. Their traces on the first three anchor columns are $100, 101, 011$, three distinct signatures, again impossible. Hence $a \ge 2$.

When $a, b \ge 2$, let $t_1 \le t_2$ be the first two row modes and $u_1 \le u_2$ the first two column modes. The ones at $(1, 2)$ and $(2, 1)$ give $t_1 = u_2$ and $t_2 = u_1$. Common mode order forces all four to agree. Distinct full traces prohibit reuse of an anchor group; order therefore gives groups 1 and 2 on both axes. An extra anchor body row with a unit signature would repeat one of those groups and thus a full row trace. An anchor in another mode has zero signature. Neither can be a body row. The identical argument applies to columns, so $a = b = 2$, and signatures identify the four body roles. This step uses full twins, not just the leading two-by-two anchor.

It remains to consider $b \le 1$. If $b = 0, a = 2$, anchor row 2 puts nonanchor columns 1 and 4 in one class. Their traces on nonanchor rows 3 and 4 are $10, 01$, impossible. If $b = 0, a \ge 3$, the positive supports of the first three anchor rows are $\{2, 3\}, \{1, 4\}, \{1, 3\}$. Their overlaps put all four columns in one class, but their traces on these three anchor rows are $011, 100, 101, 010$, four distinct signatures. This is impossible for either a variable or dummy class.

Finally, if $b = 1, a \ge 3$, the ones at $(2, 1)$ and $(3, 1)$ put anchor rows 2 and 3 in the same complementary group of the first anchor column's mode. They would be full twins, contrary to the pattern. The only noncanonical possibility is $(a, b) = (2, 1)$. □

5.2 Canonical copies: two anchor columns

For a canonical copy, the usual four-mode calculation applies with no lower bound on the anchor size. In V_i^+, let p be the child row node, a the parent row node, and b the parent column node. Row order and the bottom-right one give $\pi(p) \le a \le b$, so the top-right entry is one, impossible. In V_i^-, row order and the top-right zero give $b \le a \le \pi(p)$; the bottom-right strict threshold is zero, also impossible.

In $W_{i,d}^+$, child row and column nodes p, q have the same parity. The top-left one and column order give $\pi(p) \le b \le \pi(q)$, so $p \le q$, contradicting the top-right zero. In $W_{i,d}^-$, the inequalities instead give $\pi(q) \le b \le \pi(p)$, hence $p \ge q$. At $i < h$ the top-left entry $\mathbf{1}[p < q]$ is zero. At $i = h$ the weak leaf threshold forces $p = q$ and $b = \pi(p)$. This establishes Theorem 5.1 for canonical copies.

5.3 Unpinned copies: one anchor column

Write x, y for selected rows 3 and 4 and c_2, c_3, c_4 for the nonanchor columns. The one at $(2, 1)$ puts anchor row 2 and anchor column 1 in complementary groups of some mode t. If column 1 were group 2, then x would be in row role 2 and c_4 in column role 1 of t. Their entry is always one by Lemma 4.1, contrary to $H_4(3, 4) = 0$. Consequently column 1 is group 1, row 2 is group 2, and

$$x \in \mathsf{R}_1(t), \quad c_4 \in \mathsf{C}_2(t), \quad A_h(x, c_4) = 0. \tag{5}$$

The first anchor row has ones at c_2, c_3, putting them in one nonanchor class K. Their entries against x are $0, 1$. Thus K is variable, and x is not dummy. Also c_4 is not dummy since its entry

against x is zero. Class K differs from the class of c_4: otherwise $c_3 < c_4$ and nonanchor-row monotonicity would contradict their values $1, 0$. Let t_0 be the first anchor row's mode. Its order gives $t_0 \le t$.

Case $t = V_i^+$. Row x is child-plus node p, possibly a leaf, and c_4 is parent-plus node b. Its zero gives $\pi(p) > b$. If K is the child class, write q for c_3's node. Its one gives $p \le q$, and $c_3 < c_4$ gives $\pi(q) < b$. Thus $\pi(p) \le \pi(q) < b$, a contradiction. Only at a leaf can K alternatively be the negative parent class. Then its one gives $\pi(p) < q$, while the same-level minus-before-plus order gives $q < b$. This also contradicts $\pi(p) > b$.

Case $t = V_i^-$. Row x is a nonleaf minus node p at depth $j = i - 1$. Column c_4 is in its own minus class at node b; its zero gives $p \ge b$. The only other possible support class is the previous minus class, which does not exist when $j = 0$. Otherwise let its node be q. Its one and its order before c_4 give

$$\pi(p) < q < \pi(b) \le \pi(p),$$

a contradiction.

Case $t = W_{i,d}^+$. Row x and column c_4 are child-plus nodes p, b of the same parity, possibly shared leaves. Their zero gives $p > b$. If K is the positive parent class, its node q satisfies

$$\pi(p) \le q \le \pi(b).$$

Equal parity would imply $p \le b$, impossible. At a leaf the negative parent class is also possible in principle; it instead gives $\pi(p) < q < \pi(b)$, again impossible.

Case $t = W_{i,d}^-$, nonterminal $i < h$. Row x is minus node p of parity d, and c_4 is parent-minus node b. Its zero gives $b \le \pi(p)$. The remaining class K must be the own minus class. If its node at c_3 is q, the one and column order give

$$p < q, \qquad \pi(q) \le b \le \pi(p).$$

Parent monotonicity forces $\pi(p) = \pi(q) = b$. Binary siblings with $p < q$ must be

$$p = 2b, \qquad q = 2b + 1. \tag{6}$$

Thus $d = 0$.

Now use the anchor mode order. A positive anchor incidence with the nonterminal column class C_i^- is possible only in group 1 of $W_{i,e}^-$, with columns restricted to parity e, or in group 2 of V_{i+1}^- or $W_{i+1,e}^-$. The next-level modes come after $t = W_{i,0}^-$, so cannot supply the first anchor row. Mode $W_{i,1}^-$ also comes after t. The first row must therefore use $e = 0$, forcing q even, contrary to (6). This excludes every nonterminal exception.

Case $t = W_{h,d}^-$, terminal. Now x is leaf p, and c_4 is negative parent node b with $b \le \pi(p)$. If K is the leaf class, its node q at c_3 satisfies

$$p \le q, \qquad \pi(q) \le b \le \pi(p),$$

so $b = \pi(p) = \pi(q)$. It is unnecessary, and generally false, to force $p = q$. If K is the positive parent class, its node q gives

$$\pi(p) \le q \le b \le \pi(p),$$

so $b = q = \pi(p)$. The negative parent class was already excluded by monotonicity. These are every possible support class.

Both terminal possibilities put the fixed zero $(3, 4)$ into Z_h, completing the proof of Theorem 5.1. Each leaf row has size one and its parent-minus column block has size two, so $|Z_h| = 2m$.

Corollary 5.3 (A sufficient copy bound)**.** *For every $h \geq 1$, $N_{H_4}(A_h) \leq 2mn^6$.*

Proof. Fix the images of pattern row 3 and column 4 to be one of the $2m$ cells in Z_h. The other three row images and three column images have at most n^6 choices. Ordering and matching can only reduce this count. □

6 Distance against arbitrary edits

Theorem 6.1. *For every $h \geq 1$, every H_4-free binary $n \times n$ matrix differs from A_h in at least m^2 entries.*

Proof. Let B be any binary matrix whose changed-cell set E has $|E| < m^2$. We prove B contains H_4.

Partition each axis into its anchor groups, its dummy class, its signed variable classes below the leaves, and its shared leaf class counted once. Every class has size m. Choose a uniform leaf $z \in \{0, \ldots, m-1\}$ and let

$$a_i = \left\lfloor z/2^{h-i} \right\rfloor \quad (0 \leq i \leq h).$$

Conditionally on z, choose uniform positions

$$x_i^\sigma \in R_i^\sigma(a_i), \qquad y_i^\sigma \in C_i^\sigma(a_i)$$

independently within distinct blocks, using just one choice on each axis at the shared leaf. Independently choose a uniform position from every anchor group and both dummy blocks; call the latter x_*, y_*.

Each variable representative is uniform on its size-m class: its node has probability 2^{-i}, and each within-block choice has conditional probability $2^i/m$. An anchor or dummy representative is independent of every opposite-axis representative. The root representatives are independent uniform choices in fixed size-m blocks. Hence, for each fixed cell with at least one anchor or dummy coordinate, or with both coordinates at root level,

$$\mathbb{P}(\text{both coordinates are selected}) = m^{-2}.$$

Call these cells protected. Nonroot variable pairs may be correlated; no m^{-2} claim is made for them. There is one representative per class, including just one leaf representative per axis. Summing over the set of changed protected cells, each counted once, yields

$$\mathbb{P}(\text{some selected protected cell belongs to } E) \leq |E|/m^2 < 1.$$

Fix a selection avoiding every such change. Its anchor entries, anchor-to-body signatures, dummy incidences, and root entries agree with the original host. All remaining entries of B are unrestricted.

In each mode, the selected two anchor rows and columns form the required $\mathcal{C}_2$, preceding the selected body positions. Any correctly ordered selected body equal to $P = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ therefore extends to an actual H_4-copy in B. Suppose, toward a contradiction, that B is free. Its unchanged roots give

$$B(x_0^+, y_0^+) = 1, \qquad B(x_0^-, y_0^-) = 0.$$

Assume these values have reached level $i-1$. The selected V_i^+ and V_i^- bodies are

$$\begin{pmatrix} 1 & B(x_i^+, y_{i-1}^+) \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 1 & B(x_i^-, y_{i-1}^-) \end{pmatrix}.$$

Their orders follow from the common parent-child path and (1); the left columns are intact dummy incidences. Avoiding P forces the two unknown entries to be one and zero, respectively. Put $d = a_i \bmod 2$. The selected $W_{i,d}^+$ and $W_{i,d}^-$ bodies then are

$$\begin{pmatrix} 1 & B(x_i^+, y_i^+) \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} B(x_i^-, y_i^-) & 0 \\ 1 & 1 \end{pmatrix}.$$

Their bottom rows are intact dummy incidences, and their column orders again follow from the common path. Avoiding P forces

$$B(x_i^+, y_i^+) = 1, \qquad B(x_i^-, y_i^-) = 0.$$

Induction reaches $i = h$, where the two asserted values refer to the same shared leaf cell. This is impossible. Thus B contains a copy, proving the lower bound. □

This proof is independent of the all-copy classification in the *unedited* host. It allows arbitrary zero-to-one and one-to-zero changes in every region. It never substitutes hitting original copies for obtaining a free edited matrix. In particular, no claim is made that flipping Z_h, or the prior dummy-by-leaf rectangle, repairs the two-group host. Nor do we claim the minimum repair size equals m^2; the lower bound is what the theorem needs.

7 Complete copy classification and exact counts

Lemma 7.1. *The fourth selected row of every copy is dummy.*

Proof. Canonical copies already have terminal minus-horizontal roles, so the fourth row is dummy. In an unpinned terminal copy, x is leaf row 3, and class K of columns 2 and 3 is either the leaf class or the positive parent class. Row 4 has ones on K and the negative parent column c_4. If K is the leaf class, any such nondummy row is a leaf, since only leaf variable rows meet leaf columns. If K is the positive parent class, any variable row meeting both positive and negative parent classes must again be a leaf. A leaf row after x has a larger node index; its entry at c_2 is no greater than x's zero, contradicting the required one. Thus row 4 is dummy. □

Write $M = m/2$. The following three families are exhaustive. In each description choose arbitrary actual representatives of the displayed blocks. The multiplicity of one block description is the product of all eight block sizes. The full pattern has distinct rows and columns, so no copy can select a twin block twice.

7.1 Canonical family

For every leaf $p = 2b + d$, use both anchor groups on both axes of $W_{h,d}^-$, then leaf row p, a dummy row, leaf column p, and a column of $C_{h-1}^-(b)$. Conversely these always form a copy. There are m block descriptions, each of multiplicity $2m^5$, for

$$2m^6 \tag{7}$$

expanded copies.

7.2 Unpinned family with two positive-parent columns

Choose

$$0 \le a < b < M, \qquad d \in \{0,1\}, \qquad p = 2b + d.$$

The columns, in order, are group 1 of $W^-_{h,d}$, $C^+_{h-1}(a)$, $C^+_{h-1}(b)$, $C^-_{h-1}(b)$. The last three rows are group 2 of that minus mode, leaf p, and the dummy. The first row is precisely one of

$$\widehat{R}_{V^+_h}(2), \quad \widehat{R}_{W^+_{h,0}}(1), \quad \widehat{R}_{W^+_{h,1}}(1),$$

or, when $h \ge 2$ and $a \equiv b \equiv e \pmod 2$, $\widehat{R}_{W^+_{h-1,e}}(2)$.

To verify necessity, localization gives the parent index of c_4 as $b = \pi(p)$. The zero and one on c_2, c_3, together with their order before c_4, force their indices to be $a < b$ and b, respectively. The displayed first rows are exactly the anchor incidences with both positive-parent columns, as seen from the complete incidence list. They precede the second row. The role table checks every entry, proving sufficiency too.

Thus the number of block descriptions is

$$6\binom{M}{2} + 2\left\{\binom{\lceil M/2\rceil}{2} + \binom{\lfloor M/2\rfloor}{2}\right\}.$$

Each has multiplicity $8m^4$. For $h \ge 2$, M is even, and the expanded contribution is

$$7m^6 - 16m^5. \tag{8}$$

For $h = 1$, no pair $a < b$ exists, so this family is empty.

7.3 Unpinned family with two leaf columns

Choose $p = 2b + d$, two leaf nodes $q_0 < p \le q_1$ with $\pi(q_1) = b$, and common column parity $q_0 \equiv q_1 \equiv e \pmod 2$. The columns are group 1 of $W^-_{h,d}$, leaves q_0, q_1, and $C^-_{h-1}(b)$. The last three rows again are group 2 of that minus mode, leaf p, and dummy. The first row is $\widehat{R}_{W^+_{h,e}}(2)$, or, if $e \le d$, $\widehat{R}_{W^-_{h,e}}(1)$.

Necessity follows from the leaf threshold, localization, and the fact that these are exactly the anchor incidences with leaf columns. The inequality $e \le d$ is precisely the first-anchor-row order condition for a minus mode; all plus modes precede the minus modes. The role and threshold tables directly verify sufficiency. For each parent b, the possibilities are

p	q_1	number of descriptions
$2b+1$	$2b+1$	$2b$
$2b$	$2b$	$2b$
$2b$	$2b+1$	b

because there are b earlier leaves of the required parity, with two possible first rows in the first two cases and one in the last. Summing gives $5\binom{M}{2}$ descriptions, each of multiplicity $2m^4$, for

$$\frac{5}{4}m^6 - \frac{5}{2}m^5. \tag{9}$$

This formula is zero at $m = 2$, as required.

7.4 Conclusion and comparison of bounds

The shape lemma, all-depth mode analysis, and dummy-row lemma prove that the three descriptions are exhaustive. They are disjoint by anchor count and the class of their two middle columns. Adding (7)–(9) gives

$$N_{H_4}(A_h) = \frac{41}{4}m^6 - \frac{37}{2}m^5 \quad (h \geq 2).$$

At $h = 1$ only the canonical family remains, giving 128. In particular $N_{H_4}(A_h) \leq (41/4)m^6$ for every $h \geq 1$. Dividing by $n^8 = d_h^8 m^8$ proves

$$\frac{N_{H_4}(A_h)}{n^8} \leq \frac{41}{4} d_h^{-8} 4^{-h} = \frac{41}{4} \varepsilon_h^4 \, 4^{-h}.$$

The exact count is not necessary for the main size improvement. Corollary 5.3 already gives

$$\frac{N_{H_4}(A_h)}{n^8} \leq 2\varepsilon_h 2^{-h}.$$

For arbitrary $c, C > 0$, either bound divided by ε_h^C tends to zero: an exponential in h dominates every fixed real power of $14h + 2$. Taking h sufficiently large proves the fully quantified failure statement in Section 1, and completes Theorem 1.1.

Corollary 7.2 (Arbitrarily large orders at each fixed depth)**.** *For every $h \geq 1$ and every integer $t \geq 1$, replace each entry of A_h by a constant $t \times t$ block, with consecutive clones on each axis. The resulting matrix has order tn_h, normalized distance at least ε_h, and normalized copy count exactly $N_{H_4}(A_h)/n_h^8$.*

Proof. All partition classes now have size tm. In the protected-path proof, choose the common tree leaf uniformly from the original m leaves, then independent uniform representatives within the enlarged blocks, including the size-t leaf blocks. Every protected-cell selection probability is $(tm)^{-2}$. The same propagation contradiction proves a lower bound $(tm)^2$ on edits, and hence the same normalized distance. Two clones of one original row or column cannot occur in an H_4-copy, since its row and column traces are distinct. Every copy therefore projects to a unique original copy, and each original copy has exactly t^8 lifts. Division by $(tn_h)^8$ proves the count assertion. □

Thus the example also defeats formulations that allow an arbitrary order threshold depending on ε: first choose h to make the normalized copy density small, then choose t to exceed that threshold. No interchange of ε- and n-quantifiers is needed.

8 Adversarial order control and what has not been proved

8.1 A negative control for the parity order

The critical parity-order step is not cosmetic. Reverse $W_{i,0}^-$ and $W_{i,1}^-$ at each level on both axes, leaving every entry rule unchanged. For every $h \geq 3$, the following fixed-depth positions form an ordered copy:

$$\begin{array}{r|l} \text{rows} & \widehat{R}_{W_{2,1}^-}(1),\ \widehat{R}_{W_{2,0}^-}(2),\ R_2^-(2),\ R_* \\ \text{columns} & \widehat{C}_{W_{2,0}^-}(1),\ C_2^-(1),\ C_2^-(3),\ C_1^-(1). \end{array}$$

Their traces are $0110, 1001, 1010, 0111$. All selected variable positions are nonleaves. The multiplicity is

$$\left(m \cdot m \cdot \frac{m}{4} \cdot m\right)\left(m \cdot \frac{m}{4} \cdot \frac{m}{4} \cdot \frac{m}{2}\right) = \frac{m^8}{128}.$$

Hence the altered construction has copy density at least $1/(128d_h^8)$, and cannot satisfy our superpolynomial copy estimate. This proves failure of the localization mechanism under that alteration; it does not prove polynomial removal for the pattern in all other hosts.

8.2 Weak steps examined

1. **Unintended copies.** The anchor itself is not rigid. Lemma 5.2 exhausts all 25 pairs of total anchor counts by explicit cases and uses full twins where unit signatures alone would be inadequate.
2. **Arbitrary depth.** Nonanchor crown exclusion is proved using a support supergraph for every depth. The remaining proof has four mode types, the terminal/nonterminal distinction, and explicit parity inequalities. No depth cutoff is used.
3. **Shared leaves.** The strict minus threshold stops before the leaves, both parent signs are included for leaf rows, and the terminal unpinned case permits siblings $p \neq q$.
4. **Dummy positions.** Dummy columns are before all variable columns; dummy rows follow them. Their full twins and constant nonanchor entries are used only in their stated domains.
5. **Root boundary.** At V_1^- the previous minus class does not exist; that branch is excluded directly. For $h = 1$, there are no nonterminal minus-horizontal cases, and the exact count is treated separately.
6. **Repair versus hitting.** Sampling protects actual cells of an arbitrary edited matrix. Nonroot variable pairs can be correlated and can have arbitrary edited values. No proposed transversal is assumed to be a free repair.
7. **Novelty and formal status.** The candidate and tree construction are credited to prior work. No successful 4×4 theorem was located in the dated search, but publication priority has not been proved. The new theorem has no completed Lean formalization and has not undergone external peer review.

Three separate mathematical derivations in this investigation agreed on the localization argument. A further adversarial reading found two wording overstatements in working notes: monotonicity must be restricted to nonanchor rows, and the support graph is contained in the displayed graph rather than necessarily equal to it. Both are corrected in this report; neither changes the proof's required hypotheses. The exact three-family count was independently rederived rather than inferred from the finite table.

9 Reproducible artifacts and executed verification

9.1 Independent implementations

The archive contains standard-library Python programs:

- `code/generate.py`: emits every entry of A_h as binary text or as an exact weighted block matrix, with deterministic ordering.
- `code/discover.py`: independently written discovery implementation; recursively orders the tree, assigns entries by role predicates, inspects every row-block four-tuple, and counts all compatible column chains with integer arithmetic.
- `code/independent_verify.py`: imports neither the discovery program nor prior code. It constructs order using rational interval endpoints and constructs signatures using explicit

role sets. It inspects every *column*-block four-tuple and enumerates all compatible row chains. It verifies the entire certificate set, every one and zero, strict orders, exact multiplicities, and the localized zero coordinate.

- `code/families.py`: generates the proved three families and compares their complete sets and integer weights with the exhaustive certificates.
- `code/finite_checks.py`: checks the original and suggested 38-order anchors, all Boolean transition cases, parent-residue inequalities, and the reversed-parity negative-control witness.

No solver output, floating-point arithmetic, or randomized search is used for a proof claim. Timings alone use a wall clock.

9.2 Executed complete finite searches

h	n_h	block order	four-tuples	block copies	expanded copies
1	32	17	2,380	2	128
2	120	35	52,360	15	23,040
3	352	59	455,126	78	2,080,768
4	928	95	3,183,545	348	152,567,808
5	2,304	155	23,130,030	1,464	10,385,096,704

At each depth the discovery program inspected all row four-tuples; the verifier independently inspected all column four-tuples. Every complete weighted copy set agreed. The independent verifier completed the five-depth run in approximately 18.75 seconds in this environment. Complete per-embedding certificates, execution receipts, and a full expanded depth-one host are included.

For a certificate, an anchor descriptor is `["A", level, kind, parity, coordinate]`, with coordinates zero-based $0, 1$, kind `V+`, `V-`, `W+`, or `W-`, and null parity for V. A variable descriptor is `["T", level, node, sign]`, where `L` denotes the shared leaf. Dummy is `["D"]`. Each record has four rows, four columns, and an exact integer product weight.

Twin compression is exact: selecting two positions from one block would repeat a full row or column of the host, impossible because the four pattern traces on that axis are distinct. Thus a valid compressed copy expands into exactly the product of its eight block multiplicities, and all expanded copies arise uniquely. This is why the finite certificates count the full host, not just a smaller surrogate.

9.3 Why the finite checks do not extrapolate

The finite runs establish the stated finite copy sets and check two independent implementations. The infinite result follows from Lemmas 4.1–5.2, the exhaustive symbolic mode cases, and the protected-path induction, all quantified over arbitrary $h \geq 1$. Their finite case structure is independent of h: only signs, the four mode types, child residues, and the terminal boundary are split. Parent monotonicity follows from $p = 2\pi(p) + (p \bmod 2)$; the only residue values are $0, 1$. The exact count follows from complete symbolic families and finite sum identities, not a fitted sequence.

9.4 Commands

From the extracted archive directory:

```
python3 code/independent_verify.py --max-depth 5 \
  --output logs/recheck.json
python3 code/families.py --max-h 5
python3 code/finite_checks.py --output logs/finite_recheck.json
python3 code/generate.py --depth 2 --format txt \
  --output host_h2.txt
python3 code/discover.py --h 5 --count --collect \
  --output certificates/h4_copies_h5.json
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The first command verifies all provided certificates without modifying them. The discovery command explicitly regenerates the named one. The largest full binary host grows quadratically in n_h as a file; the JSON representation preserves weights and is usually more convenient. The programs raise explicit exceptions, so optimized Python does not disable their checks.

10 A kernel-checked Lean route

The formalization should first target the simpler sparse-zero bound; the exact count can be a subsequent theorem. No unproved Lean stub is presented as an executed result here.

1. Define $H_4 : \mathrm{Fin}\ 4 \to \mathrm{Fin}\ 4 \to \mathrm{Bool}$, ordered copies, and unrestricted Hamming distance as in the existing Model module. Prove finite distinctness of its four rows and columns with ordinary kernel-checked `decide`.

2. Parameterize the anchor-group type in the host, replacing Fin 64 by Fin 2, and use $d_h = 14h + 2$. Reuse or adapt `TreePositions`, `HostTreeOrder`, and `HostBodyOrder`. Preserve one literal leaf position type with both sign views.

3. Package structural hypotheses: full twins, a common anchor prefix and mode order, one-class positive neighborhoods, nested nonanchor traces, two signatures per variable class, and nonanchor crown exclusion. Prove them from the concrete role and entry tables.

4. Prove the abstract anchor-count lemma for increasing maps $\mathrm{Fin}\ 4 \to [n]$. Formalize its (a, b) cases explicitly. The $(2, 2)$ branch uses full pattern traces to exclude extra anchors; reusing a large-anchor density theorem would not suffice.

5. For $(2, 1)$, extract the complementary groups and roles. Prove the four all-depth mode cases using order equivalences and integer division. The nonterminal minus case should expose $p = 2b,\ q = 2b + 1$ and then use the concrete mode ordering. These local arithmetic implications are suitable for `omega` after unfolding the relevant natural-number facts.

6. Obtain an injection of copies into $Z_h \times [n]^3 \times [n]^3$, proving $\#Z_h = 2m$ and $N_{H_4}(A_h) \le 2mn^6$. The fixed coordinate is $(3, 4)$, with value zero.

7. Reuse the generic Boolean `PathPropagation` and finite `SamplingDistance` arguments. Re-establish the two-group frame extension and exact protected-cell sampling bound. Quantify over every binary edited matrix.

8. Adapt `SequenceAssembly` and `Consequence` with exponents $8, 6$ and coefficient 2. Prove the limit giving the fully quantified failure theorem.

9. Optionally formalize the three explicit copy families and their disjoint cardinalities to obtain the exact count.

The final verification gate should build the concrete solution, run the comparator on the exact new definitions and theorem, and print its axioms. Permit only the standard logical axioms already used upstream, with no `sorryAx`, new postulated combinatorial lemmas, or unchecked native-computation oracle. This is a proposed verification route, not a completed build in this investigation.

A The suggested 38-by-38 anchor is valid but superseded

For one-based $u, v \in [38]$, start from $J_{38} - I_{38}$. Replace rows $7, \ldots, 38$ and columns $34, \ldots, 38$ by the 32 lexicographically ordered five-bit words. Call this matrix S_{38}. Its first six rows and columns are unaffected by the replacement, and every opposite trace there has at most one zero. Each row has at least 32 ones, each column at least 22. Its rows and columns are pairwise distinct: rows $1, \ldots, 33$ have individual zeros in the first 33 columns; the last five rows are distinguished by different word suffixes. The first 33 columns have individual zero positions, while each word column has 16 zeros and the five coordinate functions differ.

Use the original tree host with this anchor. Each variable position has at most four mode incidences against distinct selected anchor groups. Dummy twins allow at most one selected dummy on each axis. No five nonanchor columns are shattered: if two columns lie in one variable class, nesting forbids the two opposite mixed traces. Otherwise five distinct classes would need a non-dummy row with four ones and a zero to realize all words, but a variable row meets at most three classes. A dummy column itself also prevents shattering.

If a selected S_{38}-copy has at most six anchors on each axis, its final 32 rows and last five columns are nonanchors, contradicting the word block. Otherwise it has at least six anchor positions on one axis. On its first six anchors every opposite pattern trace has at least five ones, excluding all variable positions on the other axis. All but at most one opposite position are anchors. Two such anchors have a common positive entry on the six-position core, so all belong to one mode. Density excludes anchor positions of other modes on the original axis. A remaining dummy on the opposite axis has at most one positive entry against that mode's selected groups, contradicting the dense core. Finally every nonanchor on the original axis has at most one positive entry against the complete opposite mode groups, contradicting row or column density. Full distinctness and order exhaust all 38 groups on both axes.

Thus the entire baseline argument works and gives a 40×40 pattern. The finite anchor checks in the archive confirm the exact properties above. This is not claimed as a novel size improvement, because the prior 5×5 construction is already much smaller. The successful new argument does not use this anchor.

B Dated source scope and novelty limits

The record distinguishes source availability from novelty.

- The OpenAI main commit, family-path histories, comparator, solution implementation, public PR collection, and relevant PR were inspected. No family-190 size change was present.
- The ProveIt path was checked again during the investigation. Its current main was `1703447ff43256042649da80c73f75121c48c95b`; the OrderedMatrices path still had only the October 7 intake commit. Its research directory contained the same 5×5 package, including the displayed 4×4 candidate.

- Searches covered the exact title, ordered binary matrix removal, four-by-four and five-by-five variants, the relevant repositories, and citations and follow-ups to [4]. No separate successful 4×4 theorem was located.
- The mathematical proof of a smaller obstruction is independent of this absence finding. Global publication priority is not verified. Ordinary web search initially missed the 5×5 report; the GitHub code search was materially more informative.

The accompanying audits give the reviewed URLs, dates, query scope, and the distinction between source inspection, prior-code reruns, and independent finite verification.

References

[1] OpenAI, *Polynomial removal fails for ordered binary matrices*, 25 September 2026, family 190, snapshot `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. https://github.com/openai/math/tree/fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb/preprints/Polynomial-removal-fails-for-ordered-binary-matrices-September-25-2026.

[2] OpenAI, family-190 Lean catalogue and MatrixRemoval implementation. https://github.com/openai/math/blob/fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb/lean/docs/190.md. https://github.com/openai/math/tree/fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb/lean/OAI/Combinatorics/MatrixRemoval.

[3] Research draft for the ProveIt project, *A 5×5 Obstruction to Polynomial Ordered Matrix Removal*, 7 October 2026, first path-history commit `bfbfe915186310fdf745a7a089e7dd88eeb773b1`. https://github.com/VladimirReshetnikov/ProveIt/blob/bfbfe915186310fdf745a7a089e7dd88eeb773b1/Combinatorics/OrderedMatrices/Research/ordered-matrix-removal/ordered_matrix_removal.tex. The candidate and failed-repair witness are in `sections/generalization_and_research.tex` at the same commit.

[4] N. Alon and O. Ben-Eliezer, *Efficient Removal Lemmas for Matrices*, arXiv:1609.04235v3, 12 June 2017; *Order* 37 (2020), 83–101. https://arxiv.org/abs/1609.04235.

[5] N. Alon, O. Ben-Eliezer, and E. Fischer, *Testing Hereditary Properties of Ordered Graphs and Matrices*, arXiv:1704.02367; FOCS 2017. https://arxiv.org/abs/1704.02367.

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[7] L. Gishboliner and I. Tomon, *Polynomial removal lemmas for ordered graphs*, *Combinatorial Theory* 2(3) (2022), Paper 3; arXiv:2110.03577v2. https://arxiv.org/abs/2110.03577.